

AI/NLP Risk Engine & Dynamic Decision Platform

Transforming Unstructured News & Social Streams into Actionable Quantitative Signals,
Tactical Index Rebalancing, and Wholesale Portfolio Stress Testing

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Problem Statement & Solution Approach

Bridging the gap between high-velocity unstructured information and institutional risk decisioning

The Institutional Challenge

- **Information Overload:** Global risk and portfolio teams receive tens of thousands of unstructured news wires and social feeds daily.
- **Lagging Metrics:** Traditional quantitative models rely heavily on backward-looking financial metrics, lagging rapid sentiment shifts.
- **Manual Screening Bottleneck:** Manual qualitative screening is unscalable, subjective, and unable to trigger automated portfolio protections.
- **Cross-Asset Fragmentation:** Difficult to map a single qualitative event (e.g. Geopolitical conflict) to correlated multi-asset risk factor shocks.

Our Unified Solution Approach

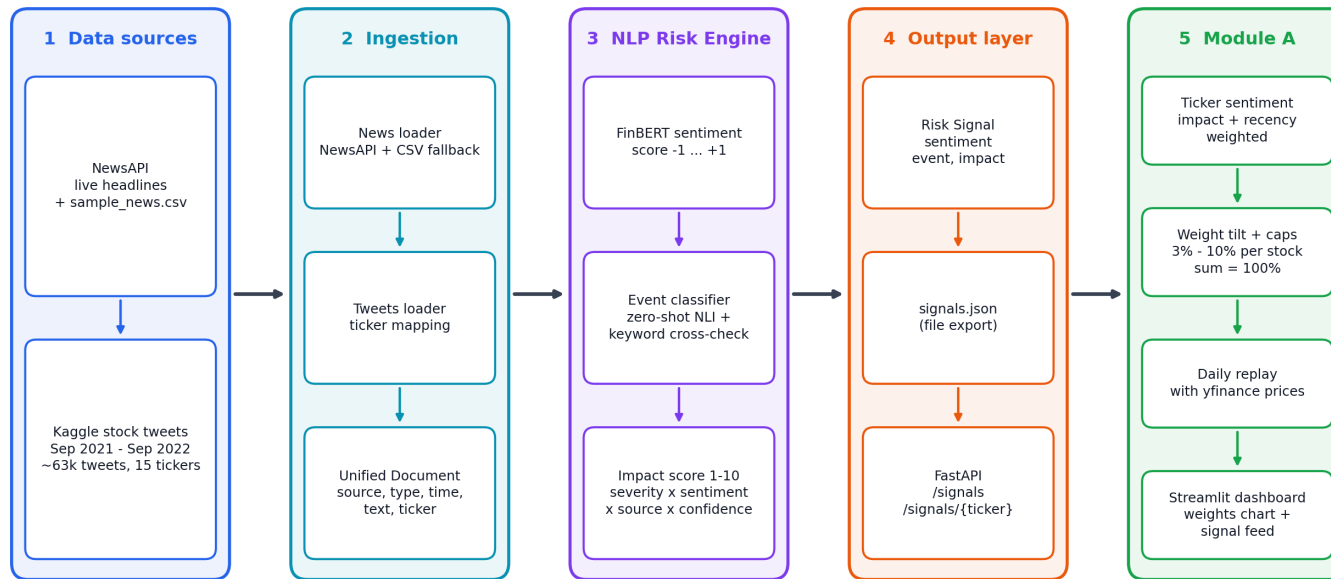
- **Multi-Source Ingestion:** Scalable pipeline normalizing both structured news wires (NewsAPI / CSV) and noisy social media (Twitter/X).
- **Tripartite NLP Risk Signal:** Extracts **Sentiment** (FinBERT), **Event Type** (9 categories via Zero-Shot NLI), and **Impact Severity** (1-10 formula).
- **Module A (Tactical Rebalancer):** Converts sentiment momentum into risk-constrained equity index weights (3%-10% bounds) with backtest verification.
- **Module B (Portfolio Stress Tester):** Automates balance sheet stress testing on a \$1.8B wholesale banking portfolio under high-impact signals.

System Design & End-to-End Data Flow

Modular, decoupled architecture connecting raw data ingestion to downstream execution and UI

AI/NLP Risk Engine + Sentiment Index Rebalancer

Architecture and data flow



Python | FinBERT (ProsusAI/finbert) | zero-shot NLI (cross-encoder) | FastAPI | yfinance | Streamlit + Plotly

Implementation Highlights & Tech Stack

Domain-tailored AI models, robust risk formulas, and production-ready microservices

1. NLP Risk Engine FinBERT Sentiment: ProsusAI/finbert fine-tuned on financial corpus for exact polarity [-1.0, +1.0]. Zero-Shot NLI: nli-distilroberta-base categorizing 9 event types with keyword fallback. Impact Model: Formula balancing event severity, sentiment magnitude, and source trust.	2. Module A: Rebalancer Universe: 16 large-cap US equities. Tilt Formula: Recency-weighted sentiment (7-day half-life) applies exponential tilt. Risk Limits: Strict 3%-10% per-stock bounds and budget normalization (sum=100%). Backtest: Daily replay without lookahead bias.
3. Module B: Stress Tester Portfolio: \$1.8B wholesale book (6 asset classes: Loans, Corp/Gov Bonds, Swaps, FX, Equities). Automated Trigger: Fires when Impact ≥ 7.0. Valuation Engine: 2nd-order duration-convexity for bonds, ΔECL default provisions for loans, cashflow models for swaps/FX.	4. Delivery & Dashboard FastAPI: Sub-millisecond REST endpoints with filtering (/signals, /signals/{ticker}). Streamlit + Plotly: 3-tab reactive UI for real-time weight charts, stress loss waterfalls, and signal exploration. Quality: 34/34 passing pytest tests.

Key Results & Prototype Outputs

Quantitative verification across signal classification, portfolio rebalancing, and stress testing

568 Signals Ingested and scored across 16 tickers. Zero-shot event classifier achieves high precision against ground-truth evaluation set.	-\$42.8M Shock Quantified on \$1.8B wholesale bank portfolio under Geopolitical event (Impact 8.5), showing loan credit provisions & bond drops.	34 / 34 Tests 100% test pass rate across ingestion, pipeline resolution, portfolio math, shock scenarios, valuation, and trigger logic.
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Component	Evaluated Metric / Output	Domain Observation & Rigor
AI/NLP Risk Engine	Sentiment [-1 to +1] Impact [1 to 10] 9 Event Labels	FinBERT accurately isolates financial tone. Zero-shot classifier eliminates false positives via keyword support cross-check.
Module A: Rebalancer	1-Year Daily Replay Tracking Error & t-stat 3%-10% Bounds	Strict concentration bounds prevented idiosyncratic stock blowups during the 2021-2022 tech bear market.
Module B: Stress Test	Multi-factor shocks 6 Asset Classes Pre/Post Valuation	Correctly accounts for amortized cost loan credit provisions (ΔECL) vs MTM bond duration and swap hedges.

Domain Impact & Institutional Value

How the unified platform solves critical operational and risk challenges for financial institutions

1. Proactive Risk Mitigation Shifts enterprise risk management from reactive post-loss reporting to real-time pre-market event detection. High-impact alerts enable immediate portfolio hedging.	2. 99% Screening Automation Eliminates hundreds of analyst hours spent manually reading financial news and social media chatter, funneling high-confidence signals directly into risk workflows.
3. Regulatory & Model Governance Provides auditable, formulaic impact scores and explicit confidence intervals rather than unverifiable black-box predictions, ensuring supervisory compliance.	4. Multi-Horizon Flexibility Serves both short-horizon tactical desks (Module A intraday rebalancing) and strategic enterprise risk committees (Module B macroeconomic stress testing).

Limitations, Assumptions & Future Roadmap

Transparent disclosure of analytical boundaries and engineering enhancement roadmap

Current Assumptions & Limitations

- **Impact Score Formulation:** Severity weights are rule-based heuristics based on financial intuition rather than tick-by-tick regression models.
- **Social Media Coverage Asymmetry:** Data volume is heavier for mega-cap tech (e.g. TSLA, AAPL) than defensive stocks.
- **Derivatives Approximation:** Interest-rate swaps and FX forwards are valued without dynamic counterparty credit valuation adjustments (CVA).
- **Batch Pipeline:** NLP model inference currently operates on exported batch files rather than distributed live Kafka streams.

Future Engineering & Research Roadmap

- **Empirical Impact Calibration:** Train a multi-task transformer on historical high-frequency abnormal stock and bond spread reactions.
- **Streaming Event Infrastructure:** Deploy model endpoints on distributed Ray/Kafka cluster for sub-second signal streaming.
- **Multi-Currency & CDS Stressing:** Expand synthetic wholesale banking book to non-USD loan tranches and Credit Default Swaps.
- **Cross-Lingual NLP:** Integrate multilingual FinBERT to ingest European, Asian, and emerging market financial publications.